

JNX-OS v2 - Master Documentation

Phase 5C Complete Reference

Project: JNX-OS v2 + Qryx AI Sales Assistant

Status: Production Ready (with Active Debugging)

Last Updated: January 2, 2026, 15:45 UTC

Version: 2.2.0 (Phase 5C - Shop Intelligence + OAuth Fix)

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Quick Start

Prerequisites

- Node.js 18+ & Yarn
- Clerk account (authentication)
- Supabase project (database)
- Stripe account (billing)
- Shopify Partner account (Qryx integration)

Environment Setup

```
# Copy from /nextjs_space/.env
NEXT_PUBLIC_CLERK_PUBLISHABLE_KEY=pk_test...
CLERK_SECRET_KEY=sk_test...

NEXT_PUBLIC_SUPABASE_URL=https://xxx.supabase.co
NEXT_PUBLIC_SUPABASE_ANON_KEY=eyJhb...
SUPABASE_SERVICE_ROLE_KEY=eyJhb...

STRIPE_SECRET_KEY=sk_live...
NEXT_PUBLIC_STRIPE_PUBLISHABLE_KEY=pk_live...
STRIPE_WEBHOOK_SECRET=whsec...

SHOPIFY_API_KEY=6e62aef5f8013048ca5b446fa86c6fae
SHOPIFY_API_SECRET=shpss...
SHOPIFY_APP_URL=https://www.jnxlabs.ai # CRITICAL!
SHOPIFY_SCOPES=read_products,read_product_listings,read_customers,read_orders

SESSION_SECRET=<base64-encoded-32-byte-key>
GEMINI_API_KEY=AIzaSy...
```

Development Server

```
cd /home/ubuntu/jnx-os/nextjs_space
yarn install
yarn dev
# Open http://localhost:3000
```

Project Overview

What is JNX-OS?

Enterprise SaaS Foundation built for multi-tenant B2B applications with:

- Clerk authentication (email/password + Google SSO)
- Supabase PostgreSQL (9 tables, 23 indexes, 6 foreign keys)
- GDPR compliance (data export, deletion, audit trails)
- Role-based access control (admin/user)
- Security by default (rate limiting, CSP, HSTS)

What is Qryx?

AI Sales Assistant for Shopify - First product on JNX-OS:

- Gemini 2.0 Flash AI integration
- Shopify OAuth integration
- Stripe billing (3 pricing tiers)
- Conversation limits based on plan
- Real-time chat widget
- **NEW: Shop Intelligence Analysis (Phase 5C)**

Tech Stack

Frontend

- **Framework:** Next.js 14.2.28 (App Router)
- **Styling:** Tailwind CSS + JNX Dark Design System
- **Animations:** Framer Motion
- **Icons:** Lucide React
- **UI Components:** Radix UI + Custom

Backend

- **Auth:** Clerk (webhooks for user sync)
- **Database:** Supabase PostgreSQL
- **Billing:** Stripe (Checkout + Webhooks)
- **AI:** Google Gemini 2.0 Flash
- **E-commerce:** Shopify OAuth

Infrastructure

- **Hosting:** Vercel (auto-deploy from GitHub)
 - **Domain:** www.jnxlabs.ai
 - **CDN:** Vercel Edge Network
 - **Environment:** Production + Preview
-

Database Schema

Core Tables (9 Total + Phase 5C Extensions)

1. `orgs` - Organizations

```
CREATE TABLE orgs (  
  org_id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),  
  name TEXT NOT NULL,  
  clerk_org_id TEXT UNIQUE,  
  created_at TIMESTAMPTZ DEFAULT NOW(),  
  updated_at TIMESTAMPTZ DEFAULT NOW()  
);
```

2. `users` - User Accounts

```
CREATE TABLE users (  
  user_id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),  
  email TEXT UNIQUE NOT NULL,  
  full_name TEXT,  
  role TEXT DEFAULT 'user',  
  org_id UUID REFERENCES orgs(org_id) ON DELETE SET NULL,  
  clerk_user_id TEXT UNIQUE NOT NULL,  
  supabase_user_id UUID,  
  deleted_at TIMESTAMPTZ,  
  created_at TIMESTAMPTZ DEFAULT NOW()  
);
```

3. billing_subscriptions - Stripe Subscriptions

```
CREATE TABLE billing_subscriptions (
  subscription_id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  clerk_user_id TEXT NOT NULL REFERENCES users(clerk_user_id) ON DELETE CASCADE,
  stripe_subscription_id TEXT UNIQUE NOT NULL,
  stripe_customer_id TEXT NOT NULL,
  plan_name TEXT NOT NULL,
  status TEXT NOT NULL,
  current_period_start TIMESTAMPTZ NOT NULL,
  current_period_end TIMESTAMPTZ NOT NULL,
  cancel_at_period_end BOOLEAN DEFAULT FALSE,
  created_at TIMESTAMPTZ DEFAULT NOW(),
  updated_at TIMESTAMPTZ DEFAULT NOW()
);
```

4. shopify_shops - Shopify Store Connections (Extended Phase 5C)

```
CREATE TABLE shopify_shops (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  org_id UUID REFERENCES orgs(org_id),
  clerk_user_id TEXT, -- PHASE 5B: Added for user-based billing
  shop_domain TEXT UNIQUE NOT NULL,
  shop_name TEXT,
  shop_email TEXT,
  shop_owner_name TEXT,
  access_token TEXT NOT NULL,
  scope TEXT,
  installed_at TIMESTAMPTZ DEFAULT NOW(),
  uninstalled_at TIMESTAMPTZ,
  plan_tier TEXT DEFAULT 'free',
  subscription_status TEXT DEFAULT 'pending',
  trial_ends_at TIMESTAMPTZ,
  shopify_charge_id TEXT,
  billing_period_start TIMESTAMPTZ,
  billing_period_end TIMESTAMPTZ,
  shopify_plan TEXT,
  country_code TEXT,
  currency TEXT,
  timezone TEXT,
  -- PHASE 5C: Shop Intelligence
  shop_intelligence JSONB, -- AI-analyzed shop profile
  analyzed_at TIMESTAMPTZ,
  deleted_at TIMESTAMPTZ,
  created_at TIMESTAMPTZ DEFAULT NOW(),
  updated_at TIMESTAMPTZ DEFAULT NOW()
);

-- Phase 5C Index for Shop Intelligence
CREATE INDEX idx_shopify_shops_intelligence ON shopify_shops USING GIN(shop_intelligence);
```

5-9. Supporting Tables

- `qryx_conversations` - Chat History
- `qryx_messages` - Individual Messages
- `audit_logs` - System Audit Trail
- `system_events` - System-Wide Events
- `data_export_requests` - GDPR Data Exports

API Endpoints

Qryx Installation Flow (CRITICAL)

Method	Endpoint	Description	Status
GET	<code>/api/qryx/install?shop=xxx</code>	Entry point - saves shop session	⚠ Debugging
GET	<code>/api/qryx/callback</code>	OAuth callback - links shop to user	✅ Fixed (SaaS flow)
POST	<code>/api/qryx/start-oauth</code>	Triggers Shopify OAuth	✅ Working
GET	<code>/api/qryx/auth</code>	Alternative OAuth entry	✅ Working

Stripe Billing

Method	Endpoint	Description
POST	<code>/api/stripe/checkout</code>	Create checkout session
POST	<code>/api/stripe/webhook</code>	Handle Stripe webhooks

Dashboard

Method	Endpoint	Auth	Description
GET	<code>/app</code>	✅	User dashboard
GET	<code>/app/qryx</code>	✅	Qryx management
GET	<code>/app/billing</code>	✅	Billing management
GET	<code>/app/products</code>	✅	Products overview
GET	<code>/admin</code>	Admin	Admin dashboard

Stripe Integration

Pricing Plans (Live Mode)

Plan	Price	Conversations	Price ID
Starter	\$29/mo	500	price_1SjkKKBQ5QFS35pBxGKE0r50
Professional	\$79/mo	2,000	price_1SjkQTBQ5QFS35pBpWkdi5ws
Business	\$199/mo	5,000	price_1Sjkr4BQ5QF535pBkhTJsxk2

Webhook Events (5)

1. checkout.session.completed
2. customer.subscription.created
3. customer.subscription.updated
4. customer.subscription.deleted
5. invoice.payment_failed

Webhook URL

<https://www.jnslabs.ai/api/stripe/webhook>

SaaS Installation Flow

Corrected 14-Step Flow (Phase 5 SaaS)

USER JOURNEY: Shopify App Installation

1. User opens installation link:
`https://www.jnslabs.ai/api/qryx/install?shop=xxx`
2. `/api/qryx/install` saves shop to session
 - └ Uses JWT-encrypted cookie (30 min TTL)
3. Check: Is user authenticated?
 - └ YES → Redirect to `/products/qryx/setup`
 - └ NO → Redirect to `/login?redirect_url=...`
4. User logs in via Clerk
5. Redirect to `/products/qryx/setup`
 - └ Plan Selection: Starter/Professional/Business
6. User selects plan → Stripe Checkout
7. After payment → Shopify OAuth triggered
 - └ `/api/qryx/start-oauth?shop=xxx`
8. Shopify OAuth Screen
 - └ User approves permissions
9. Redirect to `/api/qryx/callback`
 - └ `currentUser()` gets authenticated user
 - └ `getUserByClerkId()` gets JNX user
 - └ `upsertShopifyShop()` links shop to EXISTING org
10. Shop Intelligence Analysis (Phase 5C)
 - └ Gemini AI analyzes shop profile
 - └ Stores in `shop_intelligence JSONB`
11. Widget Installation on Shopify Store
12. Redirect to `/app/qryx` (Dashboard)

CRITICAL: Correct Installation URL

USE THIS (SaaS Flow):

`https://www.jnslabs.ai/api/qryx/install?shop=shopbotv3.myshopify.com`

NOT THIS (Old Direct OAuth):

`https://shopbotv3.myshopify.com/admin/oauth/authorize?...`

Shop Intelligence (Phase 5C)

Database Extension

```
-- Added to shopify_shops table
shop_intelligence JSONB, -- AI-analyzed shop profile
analyzed_at TIMESTAMPTZ,

-- GIN Index for JSONB queries
CREATE INDEX idx_shopify_shops_intelligence
ON shopify_shops USING GIN(shop_intelligence);
```

Shop Intelligence Schema

```
interface ShopIntelligence {
  business_type: string; // 'fashion', 'electronics', etc.
  target_audience: string; // 'millennials', 'professionals', etc.
  price_range: string; // 'budget', 'mid-range', 'luxury'
  product_categories: string[];
  brand_voice: string; // 'casual', 'professional', 'playful'
  key_selling_points: string[];
  recommended_greeting: string;
  upsell_strategies: string[];
  analyzed_at: string; // ISO timestamp
}
```

AI Integration

- **Model:** Gemini 2.0 Flash
- **Trigger:** After successful Shopify OAuth
- **Data Source:** Shop products, metadata, order history
- **Storage:** shop_intelligence JSONB column

Logo Design System

Component: components/ui/jnx-logo.tsx

```
interface JNXLogoProps {
  size?: 'sm' | 'md' | 'lg';
  variant?: 'default' | 'admin';
  animated?: boolean;
  className?: string;
}
```

Color Variants

- **Default (Cyan/Blue):** User-facing pages
- **Admin (Purple/Pink):** Admin dashboard

Integration Points

- Homepage header/footer
- QRYX product card

- User dashboard sidebar
- Admin dashboard sidebar

Testing Guide

Playwright Test Suite (24 Tests)

Scenario	Tests	Status
New User Flow	6	✓
Existing User Flow	4	✓
Session Expiry	4	✓
Payment Failure	5	✓
Webhook Retry	5	✓

Run Tests

```
cd /home/ubuntu/jnx-os/nextjs_space  
yarn playwright test
```

Test Cards (Stripe)

- **Success:** 4242 4242 4242 4242
- **Decline:** 4000 0000 0000 0002
- **3D Secure:** 4000 0025 0000 3155

Deployment Status

Current Production

- **Domain:** <https://www.jnxlabs.ai>
- **Host:** Vercel
- **Last Deploy:** January 2, 2026
- **Git Branch:** main
- **Latest Commits:**
 - `d68489d` - Debug: Add extensive logging to `/api/qryx/install`
 - `e77c40b` - Fix: Async cookies() in shop-session
 - `76f2a19` - Fix: SaaS-Flow OAuth Callback for existing users

Environment Variables (Vercel)

- ✓ `SHOPIFY_APP_URL=https://www.jnxlabs.ai`
- ✓ `SHOPIFY_API_KEY`

-  `SHOPIFY_API_SECRET`
 -  `SHOPIFY_SCOPES`
 -  All Clerk, Supabase, Stripe keys
-

Known Issues & Fixes

Issue 1: “Missing required parameters” Error

Status: FIXED (commit `76f2a19`)

Problem: Old callback route tried to create NEW Clerk users instead of using existing authenticated user.

Solution: Rewrote `/api/qryx/callback` to use `currentUser()` and link shop to existing user/org.

Issue 2: “Application error” on `/api/qryx/install`

Status: DEBUGGING (commit `d68489d`)

Problem: Server-side exception, possibly related to:

1. Async `cookies()` API in Next.js 14.2+
2. Session management
3. Authentication check

Fix Applied:

- Made `cookies()` async in `shop-session.ts`
- Added extensive logging to identify exact failure point
- Wrapped all operations in try-catch blocks

Issue 3: Shopify Redirect URL Mismatch

Status: USER ACTION REQUIRED

Problem: Shopify Partners had old redirect URL.

Fix: In Shopify Partners Dashboard:

```
Allowed redirection URL(s):
https://www.jnxxlabs.ai/api/qryx/callback
```

File Structure

```

/home/ubuntu/jnx-os/
├── nextjs_space/
│   ├── app/
│   │   ├── api/
│   │   │   ├── qryx/
│   │   │   │   ├── install/route.ts      # ⚠ Entry point (debugging)
│   │   │   │   ├── callback/route.ts     # ✅ Fixed for SaaS flow
│   │   │   │   ├── start-oauth/route.ts
│   │   │   │   ├── chat/route.ts
│   │   │   │   └── config/route.ts
│   │   │   ├── stripe/
│   │   │   │   ├── checkout/route.ts
│   │   │   │   └── webhook/route.ts
│   │   │   └── webhooks/clerk/route.ts
│   │   ├── app/
│   │   │   ├── page.tsx                  # Dashboard
│   │   │   ├── qryx/page.tsx             # Qryx management
│   │   │   ├── billing/page.tsx          # Billing (Phase 5B)
│   │   │   └── products/page.tsx
│   │   ├── products/qryx/setup/page.tsx  # Plan selection
│   │   └── admin/
│   ├── components/ui/
│   │   ├── jnx-logo.tsx                  # SVG Logo component
│   │   └── [...other components]
│   ├── lib/
│   │   ├── session/shop-session.ts       # ✅ Fixed async cookies()
│   │   ├── db/
│   │   │   ├── helpers.ts
│   │   │   ├── qryx-helpers.ts          # ✅ Added clerk_user_id support
│   │   │   └── billing-helpers.ts
│   │   ├── shopify/client.ts
│   │   ├── stripe/client.ts
│   │   ├── ai/
│   │   │   ├── gemini.ts
│   │   │   └── shop-analyzer.ts          # Phase 5C
│   └── .env
├── docs/
└── scripts/

```

Phase Roadmap

Completed

- ✅ Phase 1-3: Foundation (Auth, DB, GDPR)
- ✅ Phase 4: Qryx Core (Shopify, AI Chat)
- ✅ Phase 5A: SaaS Billing (Stripe)
- ✅ Phase 5A++: Logo Integration
- ✅ Phase 5B: Billing Dashboard UI
- ✅ Phase 5C: Shop Intelligence (DB Migration Done)

In Progress

- ⚠ Phase 5C: Testing OAuth Flow

Upcoming

- Phase 6: Enhanced Qryx Features
 - Phase 7: Scale & Optimize
-

Support

- **GitHub:** <https://github.com/JNXLabs/jnx-os>
 - **Production:** <https://www.jnxfabs.ai>
 - **Shopify Test Store:** shopbotv3.myshopify.com
 - **Email:** support@jnxfabs.ai
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